# BICYCLES AND MOTORCYCLES

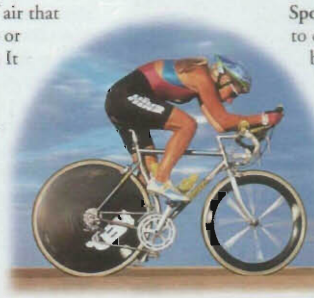
FUN, AND ENVIRONMENTALLY friendly, the bicycle is the simplest form of mechanical transport. A bicycle, or bike, is a two-wheeled machine that converts human energy into propulsion; a motorcycle, or motorbike, is a bicycle with an engine. Modern motorcycles are complex, with engine sizes ranging from 50cc (cubic centimetres) to more than 1,000cc. In many countries, such as China, most people travel or transport goods by bicycle. Across the world, bicycles and motorcycles are used for sport and leisure.

## Parts of a bicycle

From a mountain bike to a racing bike, or a hybrid (a cross between the two), all bicycles are built in a similar way. Designed to be easy to pedal and comfortable, the weight is also important, as it affects the speed at which the bike can be propelled.


Cannondale SH600, hybrid

**Reducing drag**
Drag is the resistance of air that can slow down a bicycle or motorcycle and its rider. It is reduced by creating a streamlined shape for the air to flow around – some competitive bicycle riders even shave their legs to achieve this streamlined effect.



Time-trial bike

## Parts of a motorcycle

Like a bicycle, a motorcycle has a frame, a rear wheel that drives it along, a front wheel for steering, and controls on the handlebars. Like a car, it has an internal combustion engine and suspension. The suspension supports the motorcycle's body on the wheels, and stops it being affected by the bumping of the wheels on the road.


**Motorcycle instrument panel**
Motorcycles have an instrument panel in the centre of the handlebars. Control switches for lights and indicators can be operated with hands on the handlebars.


1992 Yamaha FZR1000 Exup

**Riding a motorcycle**
A motorcycle rider changes speed by twisting the right-hand handlebar grip, and changes gear by flicking a foot lever up or down. The front brakes are operated by hand, and the rear brakes by foot. To go round a corner, the rider turns the handlebars and leans the motorcycle over.


SFX moped

**Mopeds and scooters**
Small motorcycles used for short journeys in towns and cities are called mopeds or scooters. They have small engines, so they cannot go very fast, but are very economical. Mopeds, restricted to a 50cc engine, have pedals which the rider can use on steep hills.

## Timeline

**1839** Kirkpatrick MacMillan, a Scot, invents a lever-driven bicycle.

**1863** The French Michaux brothers build the first pedal-powered bike, a velocipede.

**1868** The Michaux brothers add a steam engine to a bike, creating the first motorcycle.

**1885** In England, James Starley makes modern-style bicycles.

**1885** German Gottlieb Daimler builds an engine-powered tricycle (below).

**1901** The 1901 Werner is the first practical road-going motorcycle.

**1914–18** Motorcycles used extensively in World War I.

**1963** Dutchman Van Wijnen designs what will become the Ecocar – covered pedal-powered transport.

FIND OUT MORE: AIR • CARS AND TRUCKS • ENGINES AND MOTORS • ENERGY • FORCE AND MOTION • MACHINES, SIMPLE • MOTOR SPORTS • POLLUTION • SPORTS • TRANSPORT, HISTORY OF